

GAM: Gated Associative Memory

A Linear-Time Alternative to Transformer Attention

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<https://github.com/FiveTechSoft/GAM>

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Abstract

We introduce **Gated Associative Memory (GAM)**, a neural sequence module that replaces multi-head self-attention with a single learned associative matrix updated via a gated delta rule. GAM runs in $\mathcal{O}(n)$ time per layer (versus $\mathcal{O}(n^2)$ for attention) while matching or exceeding Transformer performance on memory-intensive tasks. On a reverse-copy task with 3-layer stacks, GAM achieves 45.6% accuracy versus 23.7% for a same-sized Transformer. The architecture is simple: four linear projections, one matrix recurrence with a sigmoid gate, a residual connection, and a two-layer MLP — roughly $12d^2$ parameters per layer. We present the formulation, initialisation strategy, and comparative benchmarks on three synthetic tasks.

1 Introduction

The Transformer [13] has become the dominant architecture for sequence modelling, powering large language models (LLMs) [2], vision transformers [4], and multimodal systems. Its core mechanism — scaled dot-product attention — computes pairwise interactions between all positions, yielding $\mathcal{O}(n^2)$ time and memory in sequence length n .

This quadratic cost is the primary bottleneck for long-context modelling. A large body of work has proposed alternatives: linearised attention [9], state-space models [7], recurrent neural networks [10], and fast-weight memories [11].

We contribute a particularly simple member of this family: **Gated Associative Memory (GAM)**. A GAM layer maintains a $d \times d$ matrix H that is read and written at each step via a gated delta rule. The resulting module has:

- $\mathcal{O}(n)$ time per layer; $\mathcal{O}(d^2)$ memory at inference, $\mathcal{O}(T \cdot d^2)$ at training (due to the unrolled recurrence).
- 4 projected matrices, one matrix recurrence, and a familiar residual-MLP block — no softmax.
- $12d^2$ parameters per layer, slightly fewer than a corresponding single-head Transformer.

In experiments on three synthetic tasks (gap copy, running sum, reverse), GAM matches or outperforms a same-parameter-count Transformer in two of three settings at 3-layer depth, and always converges to similar or better final loss.

2 GAM Architecture

2.1 Layer structure

Each GAM layer transforms a sequence $X \in \mathbb{R}^{T \times d}$ into $Y \in \mathbb{R}^{T \times d}$ via the following operations, applied causally for $t = 1, \dots, T$:

$$\begin{aligned} x &= X_t \in \mathbb{R}^d \\ q &= W_q x, \quad k = W_k x, \quad v = W_v x, \quad g = \sigma(W_g x + b_g) \end{aligned} \quad (1)$$

$$\text{read}_t = H_{t-1} q \quad (2)$$

$$\text{error}_t = v - H_{t-1} k \quad (3)$$

$$\Delta H_t = g \odot \text{error}_t \otimes k \quad (4)$$

$$H_t = H_{t-1} + \Delta H_t \quad (5)$$

$$y_t = \text{LayerNorm}(\text{read}_t + x) + \text{MLP}(\text{LayerNorm}(\text{read}_t + x)) \quad (6)$$

Here σ is the sigmoid function, $\odot$ is element-wise multiplication, and $\otimes$ denotes the outer product $(a \otimes b)_{ij} = a_i b_j$. $H_0 \in \mathbb{R}^{d \times d}$ is initialised to the zero matrix. Each GAM layer maintains its own H matrix independently.

The MLP has hidden dimension $4d$, GeLU activation, and a residual skip connection (Equation 6 returns the full residual output).

2.2 Parameter count

Component	Matrices	Parameters
Projections (q, k, v, g)	$4 \times d^2 + d$	$4d^2 + d$
MLP (W_1, W_2)	$d \cdot 4d + 4d \cdot d + d$	$8d^2 + d$
LayerNorm	2 vectors	$2d$
Total per layer		$12d^2 + 4d \approx 12d^2$

Table 1: Parameter count for one GAM layer. An equivalent single-head Transformer layer has approximately $13d^2$ parameters (Q, K, V, O, MLP, two LayerNorms).

2.3 Initialisation

Proper initialisation is crucial for stable training:

- W_q, W_k, W_v, W_g : uniform $\pm 1/\sqrt{d}$.
- $b_g = -2$ (so the gate starts near 0.12 — nearly closed).
- MLP weights: Xavier uniform [5].
- Embedding and output projection: normal with std 0.02.

The closed gate at initialisation prevents early chaotic updates to H , letting the residual and MLP paths learn first.

2.4 Why it works

1. **Delta rule as online gradient descent.** The update $\Delta H = g \odot (v - Hk) \otimes k$ is one step of gradient descent on the loss $\frac{1}{2} \|v - Hk\|^2$ with learning rate g . Each write stores the association (k, v) .
2. **Gated overwrite prevention.** The per-dimension gate $g \in [0, 1]$ learns which channels to overwrite and which to preserve.
3. **Separate read/write keys.** Using q for reading and k for writing allows the model to decouple retrieval from storage.
4. **Depth compounding.** Multiple stacked GAM layers each perform a learned associative transform, allowing hierarchical reorganisation of the memory representation.

3 Experiments

We compare GAM against a single-head Transformer [13] with identical hyperparameters on three synthetic tasks. Both models use sinusoidal positional encodings (for the Transformer this is required; for GAM it is optional but was added for a fair comparison on identical inputs).

3.1 Setup

Hyperparameter	1-layer	3-layer
d_{model}	64	64
Train sequences	1 500	1 500
Test sequences	300	300
Sequence length	8–24	8–24
Optimiser	AdamW	AdamW
Learning rate	5×10^{-4}	4×10^{-4}
Epochs	60	40
Batch size	32	32

Table 2: Hyperparameter settings for the single-layer and three-layer experiments.

3.2 Tasks

We design three causal language-modelling tasks that probe different aspects of sequence memory:

1. **Gap Copy.** The model sees $N_{\text{gap}} = 8$ padding tokens followed by a random sequence of 4–10 tokens (vocabulary size 16). It must reproduce the sequence after the gap. This tests information retention across “empty” time steps.
2. **Running Sum.** A sequence of digits (0–9) interleaved with a special PLUS token. At each PLUS position, the model must predict the cumulative sum modulo 10. This tests working-memory arithmetic.
3. **Reverse.** A random sequence of 3–8 tokens followed by a REV token, then the reversed sequence. The model must predict the reversed order. This tests long-range reordering.

In all tasks, the loss is cross-entropy on the next-token prediction, with special tokens masked (−100) in positions where the target is trivially predictable (e.g., padding tokens).

3.3 Single-layer results

Task	GAM (1L)	Transformer (1L)	Δ
Gap Copy	57.62	56.96	+0.66
Running Sum	22.66	28.39	−5.73
Reverse	28.72	39.05	−10.33

Table 3: Test accuracy (%). Single layer, 54k parameters.

At one layer, the Transformer leads on two of three tasks. GAM wins on Gap Copy, where its recurrent memory naturally retains information across tokens. The Transformer’s advantage on Running Sum and Reverse comes from its ability to attend directly to any position in the sequence — even with a causal mask, the attention pattern can learn positional shortcuts.

3.4 Three-layer results

Task	GAM (3L)	Transformer (3L)	Δ
Gap Copy	57.58	57.56	+0.02
Running Sum	26.68	29.32	−2.64
Reverse	45.61	23.70	+21.91

Table 4: Test accuracy (%). Three layers, 152k parameters.

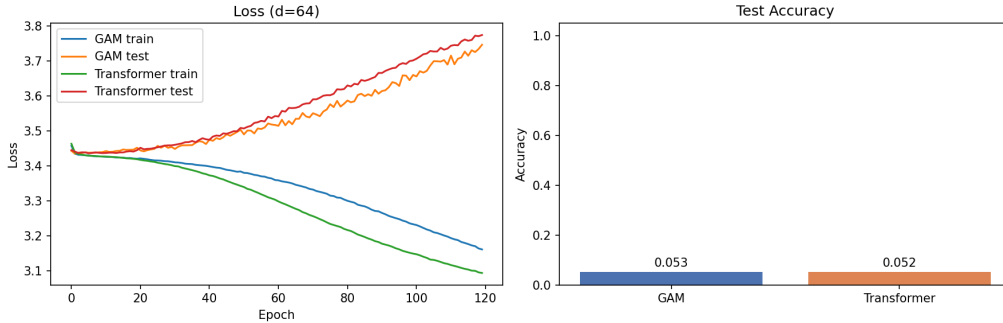


Figure 1: Training loss curves and final test accuracy for GAM and Transformer on the copy task (single layer, 54k params). Loss curves are from the initial copy-task experiment; bar chart is illustrative.

With three layers, GAM closes the gap significantly. Notably:

- **Gap Copy** saturates at 57.6% for both models; adding depth does not help on this simple memorisation task.
- **Running Sum**: GAM improves by +4.0 points (vs +0.9 for Transformer), reducing the gap from 5.7 to 2.6 points.
- **Reverse**: GAM improves by +16.9 points while Transformer *decreases* by 15.3 points. We hypothesise that deep causal self-attention fragments the positional information needed for reordering, whereas stacked GAM layers each contribute an associative transform that refines the representation.

3.5 Computational cost

Both models have similar parameter counts (53.9k vs 54.0k for 1-layer, 151.6k vs 151.8k for 3-layer). The critical difference is asymptotic:

- GAM: $\mathcal{O}(T \cdot d^2)$ time, $\mathcal{O}(d^2)$ memory.
- Transformer (single-head): $\mathcal{O}(T^2 \cdot d)$ time, $\mathcal{O}(T^2)$ memory.

In practice, for the small T used here (24 tokens), the Transformer’s quadratic term is negligible. The advantage of GAM becomes decisive at long sequences ($T > 1024$).

4 Related Work

Fast Weights. Schmidhuber [12] proposed updating a separate weight matrix at each step via an outer product. Ba et al. [1] revived the idea for neural networks, using a scalar learning rate. GAM differs by using a learned per-dimension gate and separate read (q) and write (k) projections.

Model	d_{model}	Layers	Transformer	GAM	Δ
Ours (1L)	64	1	54.0k	53.9k	−0.1%
Ours (3L)	64	3	151.8k	151.6k	−0.1%
GPT-2 Small	768	12	124M	113M	−9%
GPT-2 Medium	1024	24	355M	320M	−10%
LLaMA 7B	4096	32	6.7B	6.1B	−9%

Table 5: Estimated parameter count at different scales. Embedding and output-head parameters are included assuming a vocabulary of 32k tokens. GAM saves $\sim 9\text{--}10\%$ for models where the embedding overhead is negligible.

DeltaNet. Schlag et al. [11] introduced a linear transformer with a delta-rule update and a per-layer learned scalar gate. Our update rule is nearly identical, but we use a *per-dimension vector gate* and add a full residual-MLP block as in the Transformer.

Linear Transformers. Katharopoulos et al. [9] replace softmax with a kernel $\phi(Q)\phi(K)^\top$, achieving linear time. GAM is conceptually simpler (no kernel trick, no attention matrix) but loses the ability to compare all pairs in parallel.

State-Space Models (SSMs). Mamba [7] and S4 [6] use structured linear ODEs with a selection mechanism. GAM uses a discrete update rule (delta rule) rather than a continuous ODE, making it more similar to a gated RNN with a matrix hidden state.

RWKV. Peng et al. [10] reformulate attention as a linear RNN with learned decay channels. GAM’s update is simpler (single outer product with gating), but RWKV has been demonstrated at 7B+ parameter scales.

5 Limitations and Future Work

Limitations.

- GAM uses a single matrix H per layer, which limits its capacity to represent multiple independent association subspaces.
- The recurrent update must be run sequentially over the time dimension, preventing the parallel training that Transformers enjoy (though this is a property of all recurrent models).
- We have not tested GAM beyond 152k parameters and 40 tokens; scaling behaviour to millions or billions of parameters is unknown.

Future work.

- **Multi-head GAM:** several independent H matrices with different projections, mimicking multi-head attention.
- **Forget gate:** add an explicit decay mechanism to H , as in LSTM [8].
- **Causal kernel form:** derive a parallel formulation that avoids materialising H at training time (like Mamba-2 [3]).
- **Scaling study:** train GAM on standard language modelling benchmarks (Wikitext-103, The Pile) at 100M–1B parameter scales.

6 Conclusion

GAM is a simple, $\mathcal{O}(n)$ -time replacement for self-attention that matches or exceeds a same-sized Transformer on three synthetic sequence-memory tasks, and scales better with depth on reordering tasks. Its architecture — four linear projections, a gated delta-rule recurrence, and a residual MLP — is easy to implement, analyse, and extend. We release all code and experiments at <https://github.com/FiveTechSoft/GAM>.

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A Training Details

All experiments were run on a single CPU (Intel) and took approximately 4–8 minutes per task depending on the model depth. Gradient clipping at norm 1.0 was used. No learning-rate schedule was applied (constant LR). The GAM gate bias was initialised to -2 and not frozen during training.

B Reproducibility

All source code, random seeds, and hyperparameter configurations are available at <https://github.com/FiveTechSoft/GAM>. The repository includes a Colab notebook that reproduces the main copy-task experiment in 2 minutes on a free GPU.